

M1 Data / Input Management

complete

- ADNI rs-fMRI (BOLD only): 175 audited, 2 excluded; cohort with stage labels
- clinical tables cleaned into covariates C (LMCI: APOE $\epsilon 4$ only)
→ cohort and covariates C

M2 Preprocessing Pipeline

complete: 165 of 173 eligible scans

- drop 5 volumes; motion correction
 - EPI directly to the MNI template (no T1w)
 - resample 4 mm, smooth 6 mm
 - 27 regressors, band-pass 0.01–0.10 Hz, quality control
- 165 preprocessed scans

M3 Feature Extraction

complete: 165 of 165 scans

- 246 Brainnetome regions: time series 135×246
- ALFF, ReHo, DC per region; FC 246×246 ; normalised within subject
→ 246×3 matrix (mALFF, mReHo, DC_z) and Fisher- z FC

M4 Proposed Deep Learning Model

design only

four tasks

ALFF
ReHo
DC
FC

**Multi-attention
Transformer encoder f_θ**

246 region tokens

global attention

local attention

network attention

meta-learning loop over the four tasks

biomarker-
invariant

Z'

fusion

$H =$

$[Z' \parallel C]$

clinical covariates C
(from M1)

features

M5 Output Classifier

design only

- Bayesian ordinal head
- six ordered stages, thresholds $\theta_0 < \dots < \theta_4$
- weights $w \sim q(w)$

cumulative link:

$$P(y \leq k) = \sigma(\theta_k - g(H))$$

out: stage $\hat{y}$, class probabilities, uncertainty

M6 Dashboard

design only

- patient summary, progression chart
- class probabilities
- explainability overlays
- network view (DMN, SN, ECN)

Decision support only; reads the outputs of M1, M3, M4 and M5.



complete and verified



design only: nothing is trained and no result exists



clinical input